

Gamma distribution

Generalized form of Gamma distribution:
(Gamma distⁿ with two parameters)

A continuous R.V. X , having following P.d.F is said to follow generalized gamma distribution with two parameters ' a ' & ' λ ' :-

$$f(x) = \frac{e^{-ax} x^{\lambda-1} a^{\lambda}}{\Gamma(\lambda)}, \quad a > 0, \lambda > 0, 0 < x < \infty$$

ie $X \sim \Gamma(a, \lambda)$

Note: M.G.F = $(1 - \frac{t}{a})^{-\lambda}$

Mean = $\frac{\lambda}{a}$

Variance = $\frac{\lambda}{a^2}$

Gamma Distribution for one parameter

Let X be a continuous R.V., is said to follow a gamma distribution with parameter λ if its probability density function is given by

$$f(x) = \begin{cases} \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)}, & \lambda > 0, 0 < x < \infty \\ 0, & \text{o.w.} \end{cases}$$

Ex: Gamma distribution with single parameter $\lambda > 0$ ie $[X \sim V(\lambda)]$
when

$$\Gamma(\lambda) = \int_0^{\infty} e^{-x} x^{\lambda-1} dx, \text{ for } \lambda > 0$$

Moment Generation function of Gamma distribution:-

$$M_X(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} \cdot f(x) dx$$

$$= \int_0^{\infty} \frac{e^{tx} \cdot e^{-x} x^{\lambda-1}}{\Gamma(\lambda)} dx$$

$$= \frac{1}{\Gamma(\lambda)} \int_0^{\infty} e^{-(1-t)x} x^{\lambda-1} dx$$

$$= \frac{1}{\Gamma(\lambda)} \int_0^{\infty} \frac{e^{-z} z^{\lambda-1}}{(1-t)^{\lambda}} dz$$

$$\begin{aligned} (1-t)x &= z \\ dx &= \frac{dz}{1-t} \end{aligned}$$

$$= \frac{1}{(1-t)^{\lambda}} \cdot \frac{1}{\Gamma(\lambda)} \int_0^{\infty} e^{-z} z^{\lambda-1} dz$$

$$= \frac{1}{(1-t)^{\lambda}} \cdot \frac{1}{\Gamma(\lambda)} \Gamma(\lambda) = \frac{1}{(1-t)^{\lambda}}$$

Moment generating function = $(1-t)^{-\lambda}$

$$M_x(t) = 1 + \lambda t + \lambda^2 t^2 + \dots \infty$$

$$\boxed{\text{Mean} = \lambda}$$

$$\boxed{\text{Variance} = \lambda}$$

Note

Exponential distribution is a particular case of gamma distribution ~~is~~ (for $\lambda=1$)
if $\lambda=1$

$$f(x) = a e^{-ax} \frac{x^{1-1}}{\Gamma} , x > 0$$

$$\boxed{f(x) = a e^{-ax}}$$

which is P.d.f of exponential distribution.

is ~~xxx~~

Note: If $a=1$
 $f(x) = \frac{e^{-x} x^{\lambda-1}}{\Gamma \lambda}, x > 0$

which is gamma distⁿ with single parameter.

Both Gamma and Beta distribution are used to predict the wait time untill future event occure. for example

(a) The time between arrivals at an airport or train station

(b) the amount of time untill an equipment failure.

The difference b/w gamma distribution and exponential distribution is that the exponential distribution predicts the wait time untill the very first event. In contrast the gamma distribution indicates the wait time untill the kth event.

Properties of Gamma function:-

$$\textcircled{1} \int_0^{\infty} e^{-x} x^{\lambda-1} dx = \Gamma \lambda \quad \textcircled{2} \Gamma \lambda + 1 = \lambda \Gamma \lambda, \lambda > 0$$

$$\textcircled{3} \int_0^{\infty} e^{-ax} x^{\lambda-1} dx = \frac{\Gamma \lambda}{a^{\lambda}} \quad \textcircled{4} \Gamma \lambda + 1 = \lambda!$$

$$\textcircled{5} \Gamma 1 = 1 \quad \textcircled{6} \Gamma \frac{1}{2} = \sqrt{\pi}$$

$$\textcircled{7} \Gamma \frac{7}{2} = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{15}{8} \sqrt{\pi}$$

Q:- In a certain city, the daily consumption of electric power in million of kilowatt hours can be treated as a random variable having gamma distribution with parameters $(\frac{1}{2}, 3)$. If the power plant of this city has a daily capacity of 12 millions kilowatt hours. What is the probability that this power supply will be inadequate on any given day.

Soln:- X is a random variable, represents the daily consumption of electric power in millions of kilowatt hours

P.d.f is given by

$$f(x) = \frac{e^{-ax} a^{\lambda} x^{\lambda-1}}{\Gamma \lambda}, \quad a > 0, \lambda > 0$$

$$0 < x < \infty$$

$$a = \frac{1}{2}, \lambda = 3$$

$$f(x) = \frac{e^{-x/2} x^2 (\frac{1}{2})^3}{\Gamma 3}$$

$$f(x) = \left(\frac{1}{2}\right)^4 e^{-x/2} x^2 \quad \Gamma 3 = 2!$$

The power supply will be inadequate if $x > 12$,
Since daily capacity is of 12 millions

$$P(x > 12) = \int_{12}^{\infty} \frac{e^{-x/2} x^2}{2^4} dx$$

$$= \frac{1}{16} \left[\frac{x^2 e^{-x/2}}{-1/2} - \frac{2x e^{-x/2}}{1/4} + 2 \frac{e^{-x/2}}{-1/8} \right]_{12}^{\infty}$$

$$= \frac{1}{16} \left[(0-0+0) - \left(\frac{12^2 e^{-6}}{-\frac{1}{2}} - 8 \times 12 e^{-6} - 16 e^{-6} \right) \right]$$

$$= \frac{e^{-6}}{16} [288 + 96 + 16] = .0625$$

$$\boxed{P(X > 12) = .0625}$$

Q:- In a biomedical study with rats, a dose response investigation is used to determine the effect of the dose of a toxicant on their survival time. The toxicant is a poison discharged into the atmosphere from jet fuel. For a certain dose of toxicant the study determine the survival time (in weeks) has a gamma distribution with ~~para~~ and ~~parameters~~ $(\frac{1}{10}, 5)$. what is probability that rat survives not more than 60 weeks.

Solⁿ:- X is a random variable denotes survival time of rat (in weeks)

$$X \sim \left(\frac{1}{10}, 5 \right)$$

The P.d.f is $f(x) = \frac{e^{-ax} x^{a-1} a^a}{\Gamma(a)}$

$$f(x) = \frac{e^{-\frac{x}{10}} x^4}{\Gamma(5)} \left(\frac{1}{10} \right)^5$$

$$f(x) = \frac{e^{-\frac{x}{10}} x^4}{4! \times 10^5}; \quad x > 0$$

Probability that rat survives not more than 60 weeks = $P(X < 60)$

$$P(X < 60) = \int_0^{60} \frac{x^4 e^{-x/10}}{10^5 \times 4!} dx$$

$$= \frac{1}{10^5 \times 24} \left[\frac{x^4 e^{-x/10}}{-1/10} - \frac{4x^3 e^{-x/10}}{-1/10} + \frac{12x^2 e^{-x/10}}{-1/10} - \frac{24x e^{-x/10}}{-1/10} + \frac{24 e^{-x/10}}{-1/10} \right]_0^{60}$$

$$= \frac{e^{-6}}{10^5 \times 24} \left[0 - \left(\frac{(60)^4}{-10^{-1}} - \frac{4(60)^3}{10^{-2}} + \frac{12(60)^2}{10^{-3}} - \frac{24 \times 60}{10^{-4}} + \frac{24 \times e^{-6}}{10^{-5}} \right) \right]$$

$$= \frac{e^{-6}}{10^5 \times 24} = 0.715$$

Q:- In a certain city, the daily consumption of electric power in millions of kilowatt-hours is a random variable X having gamma distⁿ with mean 6 and variance 12.

(a) find parameters of gamma distⁿ.

(b) find the probability that on any given day the daily consumption will exceeds 12 million kilowatt hours

Solⁿ Mean = 6 variance = 12

Let X is r.v with parameter λ and a

$$X \sim \gamma(a, \lambda)$$

$$6a = \lambda$$

$$\text{i.e. } \frac{\lambda}{a} = 6,$$

$$\frac{\lambda}{a^2} = 12$$

$$12a^2 = \lambda$$

$$\text{ie } 6a = 12a^2$$

$$\text{ie } \boxed{a = \frac{1}{2}}$$

$$\& 6a = \lambda$$

$$\text{ie } \boxed{\lambda = 3}$$

Q1:- In a certain city the daily consumption of water (in millions of litres) follow approximately a gamma distribution with parameters $(\frac{1}{3}, 2)$. If the daily consumption capacity of that city is 9 millions litres of water, what is the prob. that on any given day the water supply is inadequate.

$$\text{Soln:- } f(x) = \frac{e^{-ax} x^{\lambda-1} a^{\lambda}}{\Gamma \lambda}, \quad x > 0$$

$$\text{given } a = \frac{1}{3}, \quad \lambda = 2$$

$$f(x) = \frac{e^{-\frac{1}{3}x} x^{2-1} \left(\frac{1}{3}\right)^2}{\Gamma 2}$$

$$f(x) = \frac{1}{9} x e^{-x/3}$$

$$\Gamma 2 = 1!$$

x is the r.v denotes the daily consumption of water (in millions of litres).

ie prob that on any given day, the water supply is inadequate is

$$P(X \geq 9) = 1 - P(X \leq 9)$$

$$= 1 - \int_0^9 \frac{1}{9} x e^{-x/3} dx$$

$$= 1 - \frac{1}{9} \left[x e^{-x/3} \cdot \frac{-1}{-1/3} - \frac{e^{-x/3}}{1/9} \right]_0^9$$

$$= 1 - \frac{1}{9} \left[-3x e^{-x/3} + 9 e^{-x/3} \right]_0^9$$

$$= 1 + \frac{3}{9} \left[x e^{-x/3} + 3 e^{-x/3} \right]_0^9$$

$$= 1 + \frac{1}{3} \left[(9 \times e^{-3} + 3e^{-3}) - (0 + 3e^0) \right]$$

$$= 1 + \frac{1}{3} \left[(9e^{-3} + 3e^{-3} - 3) \right]$$

$$= 1 - 1 + \frac{1}{3} \times 12e^{-3}$$

$$= 4e^{-3} = 0.1991$$

Ans